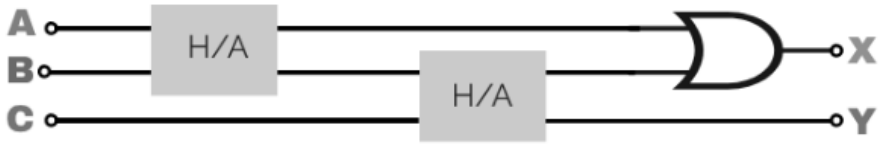
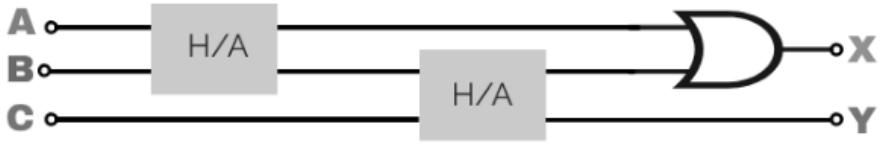
Tutorial 09 – Half Adder & Full Adder

1. How many inputs and outputs does a full adder have?
2. How can a full adder be realized in terms of other adders and gates?
3. What is the logic network shown in the figure below?



4. What are the outputs for x and y in the figure below?



5. Which of the following is known as the half adder?
    A. XOR Gate
    B. XNOR Gate
    C. NAND Gate
    D. NOT Gate
6. In which operation carry is obtained?
    A. Subtraction
    B. Addition
    C. Multiplication
    D. Both addition and subtraction
7. If A and B are the inputs of a half adder, the sum is given by _____
    A. A AND B
    B. A OR B
    C. A XOR B
    D. A EX-NOR B
8. If A and B are the inputs of a half adder, the carry is given by _____
    A. A AND B
    B. A OR B
    C. A XOR B
    D. A EX-NOR B

9. Half-adders have a major limitation in that they cannot _____
   A. Accept a carry bit from a present stage
   B. Accept a carry bit from a next stage
   C. Accept a carry bit from a previous stage
   D. Accept a carry bit from the following stages

10. The difference between half adder and full adder is _____
   A. Half adder has two inputs while full adder has four inputs
   B. Half adder has one output while full adder has two outputs
   C. Half adder has two inputs while full adder has three inputs
   D.F All of the Mentioned

11. If A, B and C are the inputs of a full adder then the sum is given by _____
   a) A AND B AND C
   b) A OR B AND C
   c) A XOR B XOR C
   d) A OR B OR C

12. If A, B and C are the inputs of a full adder then the carry is given by _____
   a) A AND B OR (A OR B) AND C
   b) A OR B OR (A AND B) C
   c) (A AND B) OR (A AND B)C
   d) A XOR B XOR (A XOR B) AND C

13. How many AND, OR and EXOR gates are required for the configuration of full adder?
   a) 1, 2, 2
   b) 2, 1, 2
   c) 3, 1, 2
   d) 4, 0, 1